



DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2021-2022
HALF YEARLY EXAMINATION (ONLINE)

CLASS: IX
SUBJECT: COMPUTER APPLICATIONS (PAPER 2)

FULL MARKS: 40
TIME: 30 minutes

Instructions:

- Attempt all questions.
 - This paper consists of ten printed pages.
 - All working including rough work must be clearly shown on the same sheet as therest of the answers.
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Question 1

a) A constant which gives the exact representation of data is called: [1]

- i) Variable
- ii) Literal
- iii) Identifier
- iv) Character

Answer: (ii)

b) Name the type of error in each case [2]

- 1) Division by a variable that contains a value of zero.
- 2) Multiplication operator used when the operation should be division.
 - i) Runtime, Logical
 - ii) Runtime, Compile time
 - iii) Logical, Compile time
 - iv) Compile time, Runtime

Answer: (i)

c) Predict the output: [2]

`System.out.println(Math.sqrt(Math.min(42.5,42.25)));`

- i) 6.0

ii) 7.0

iii) 6.5

iv) 7.5

Answer: (iii)

Explanation

`Math.min(42.5,42.25)` gives 42.25. `Math.sqrt(42.25)` gives output as 6.5.

d) An if-else construct accomplishes 'fall through'. [1]

i) True

ii) False

Answer: (ii)

e) A java statement to import *citrus* class from *Fruit* package where the Fruit package is in the *Eatables* package. [2]

i) `Import java.Fruit.Eatables`

ii) `import java.Eatables.Fruit.citrus;`

iii) `import Eatables.Fruit.citrus.*;`

iv) `import java.Eatables.Fruit.citrus.*;`

Answer: (ii)

f) Write the equivalent java expression: [1]

$$q = 1 / \sqrt{(a + b) + 3 / c^2}$$

i) `q=1/Math.sqrt(a+b)+3/c`

ii) `q=1/Math.sqrt (a)+Math.sqrt(b)+3/c*c;`

iii) `q=1/Math.sqrt(a+b)+3/Math.pow(c)`

iv) `q=1/Math.sqrt(a+b)+3/Math.pow(c,2);`

Answer: (iv)

g) Write the syntax for the functions: [1]

To find the absolute value of a number 'y'

i) Math.abs(y)

ii) Math.exp(y)

iii) y=Math.abs();

iv) Math.aps(y);

Answer: (i)

h) Predict output [2]

```
int num=20;  
switch(num)  
{  
case 10: System.out.println("TEN"); break;  
case 20: System.out.println("TWENTY"); break;  
case 30: System.out.println("THIRTY");  
}
```

i TEN

ii) TWENTY

iii) THIRTY

iv) TEN TWENTY

i) int i;

for(i = 5; i > 10; i++)

System.out.println(i);

System.out.println(i * 4);

i) 0

ii) 20

iii) 5

iv) 10

Answer: (ii)

j) x = 1; y = 1; [2]

if(n>0)

```

{   x = x + 1;
    y = y + 1; }

```

What will be the value of x and y, if n assumes a value of -3?

- i) 2,2
- ii) 1,1
- iii) 0,0
- iv) none of the above

Answer (ii)

k) Predict the output:

[2]

```

x = 5; y = 50;
while(x<=y)
{   y = y / x;
    System.out.print(y); }

```

- i) 102
- ii) 210
- iii) 012
- iv) 120

Answer: (i)

l) Predict the value of 'c'

[2]

```

int a=6,b=5,c;
c = (a++ % b++) *a + ++a*b++;

```

- i) 54
- ii) 56
- iii) 55
- iv) 53

Answer (iii)

m) Predict the number of times the loop runs:

[2]

```

class Test

```

```

{ public static void main( )
  { int i;
    for(i=0;i<5;i++)
      System.out.println(i*i*i);  } }

```

i) 4

ii) 5

iii) 6

iv) 3

Answer: (ii)

n) *If((p>q)&&(q>r)) then* [1]

(where p, q and r are three integer numbers)

i) **q is the smallest number**

ii) **q is the greatest number**

iii) **p is the greatest number**

iv) **none**

Answer (iii)

o) A compound statement can be stated as [1]

i) **p=in.nextInt();**

q=in.nextInt();

ii) **m=++a;**

n=--b;

iii) **if(a>b)**

{a++;b--;}

iv) **none**

Answer: (iii)

p) What is the difference between a While and a Do-While loop in java? [1]

i) **WHILE loop executes the statements inside of it at least once even if the condition is false.**

ii) **DO-WHILE loop executes the statements inside of it at least once even if the condition is false.**

iii) **WHILE loop is fast.**

iv) **DO-WHILE loop is fast.**

Answer: (ii)

Explanation: Both the while and do while loop work at the same speed. A do while loop executes the statements inside of it even when the condition is false. It is the reason why a do while loop is used in menu driven console java programs.

q) Predict the output:

[2]

```
for(int j=0;j<5;j++;)
System.out.print(j + “,”);
```

- i) 1,2,3,4,
- ii) 0,1,2,3,4
- iii) Compiler error
- iv) None

Answer: (iii)

Explanation:

The semicolon after the INCREMENT/DECREMENT part is not allowed.

r) Predict the output:

[2]

```
int score=1;
for(; true; score++)
{ System.out.print(score +",");
  if(score > 3)
    break;
}
```

- i) 1,2,3,
- ii) 1,2,3
- iii) 1,2,3,4,
- iv) 1,2,3,4

Answer: (iii)

Explanation: break condition is checked after printing the variable score. So it prints 4 also.

s) To execute a loop 10 times, which of the following statement satisfies:

[1]

- i) for(i=6;i<=26;i=i+2)
- ii) for(i=3;i<=30;i=i+3)
- iii) for(i=0;i<10;i=i++)
- iv) all of the above

Answer (ii)

t) The ASCII codes of upper case alphabets range from: [1]

- i) 65 - 90**
- ii) 60 - 85**
- iii) 65 - 91**
- iv) 97 - 122**

Answer (i)

u) Which of the following is a correct representation? [1]

- i) boolean m=true**
- ii) boolean m='true'**
- iii) boolean m="true"**
- iv) none**

Answer (i)

v) If int a = 25, b = 5, c = 0; what value is stored in c? When c = a % b; [1]

- i) 5.0**
- ii) 5**
- iii) 0**
- iv) none**

Answer: (iii)

w) Rewrite the following using ternary operator: [1]

```
if(a>b)  
c=a;  
else  
c=b;
```

- i) c = (b>a)?a:b;**
- ii) c = (a!=b)?a:b;**
- iii) c = (a>b)?b:a;**
- iv) None**

Answer (iv)

x) Predict the output:

[2]

```
public class Test {  
    public static void main(String[] args) {  
        int count = 1;  
        while (count <= 15) {  
            System.out.println(count % 2 == 1 ? "***" : "++++");  
            ++count;  
        }  
    }  
}
```

- i) 15 times ***
- ii) 15 times +++++
- iii) 8 times *** and 7 times +++++
- iv) Both will print only once

Answer: (iii)

Explanation: In the above code, we have declared count = 1. The value of count will be increased till 14 because of the while (count<=15) statement. If the remainder is equal to 1 on dividing the count by 2, it will print (***) else print (++++). Therefore, for all odd numbers till 15 (1, 3, 5, 7, 9, 11, 13, 15), it will print (***), and for all even numbers till 14 (2, 4, 6, 8, 10, 12, 14) it will print (++++).

Hence, an asterisk (***) will be printed eight times, and plus (+++++) will be printed seven times.

- y) Prime number is a number that is greater than 1 and divided by 1 or itself only. In other words, prime numbers can't be divided by other numbers than itself or 1. For example 2, 3, 5, 7, 11, 13, 17.... are the prime numbers.

The following programs checks if the number 'n' is prime or not. Fill the gaps from 1 to 4 to implement the program.

[4]

```
public class PrimeExample{  
    public static void main(String args[]){  
        int i,m=0,flag=0;  
        int n=3;//it is the number to be checked  
        m=__1__;  
        if(n==0||n==1){
```



```

    System.out.println(n+" is not prime number");
}else{
    for(i=2;i<=m;i++){
        if(__2__){
            System.out.println(n+" is not prime number");
            flag=1;
            __3__;
        }
    }
    if(__4__) { System.out.println(n+" is prime number"); }
} //end of else
}
}

```

i) $n/2$, $n \% i == 0$, break, flag==0

ii) $n/2$, $n \% i == 0$, break, flag==1

iii) $n/2$, $n/i == 0$, break, flag==0

iv) $n/2$, $n \% i == 0$, continue, flag==0

Answer (i)

z) Predict output:

```

int persons=45;

int random=45;

switch(45)

{case persons: System.out.print("CRICKET");

default: System.out.println("RUGBY");

}

```

i) CRICKET

ii) CRICKET RUGBY

iii) RUGBY

iv) Compiler Error

Answer (iv)

Error: case expressions must be constant expressions

So, make the variable final.

final int persons = 45;

//Then, output will be

CRICKET